

Bitcoin Equity Linkage Before vs. After 2019: Cross Period Evidence and Partial Correlation Interpretation

Chong Han

October 6, 2025

Overview and Measurement

We study how a BTC shock priced in local currency (LCU) loads into local equities, before and after 2019. Following the main paper, the crypto leg in LCU is

$$r_{i,t}^k \equiv \Delta \log \text{BTC}_{i,t}^{\text{LCU}} = \Delta \log \text{BTC}_t^{\text{USD}} + \Delta \log S_{i,t},$$

where $S_{i,t}$ is LCU per USD. For each market i , we run daily country regressions of LCU equity returns on the LCU BTC leg with progressively richer controls:

$$r_{i,t}^s = \alpha_i + \beta_i r_{i,t}^k + \Gamma_i' X_t + \varepsilon_{i,t},$$

where X_t includes $\Delta \text{Commodity}$, ΔDollar (DXY), ΔVIX , short/long rate changes, and (where applicable) slope/spread terms. We focus on the *fully controlled* specification in each period as the best analog of a *partial* (ceteris-paribus) equity–BTC link.

Interpreting the BTC coefficient as a partial effect. If y and x are standardized and the other regressors are partialled out, the OLS slope on x equals the *partial correlation* $\rho(y, x | X)$. Our daily regressions are not standardized, but the sign and relative magnitude of the fully controlled $\hat{\beta}_i$ across periods provide the same economic information: how strongly equities co-move with BTC *holding fixed* global macro shocks.

Executive Summary (Pre-2019 → Post-2019)

- **BTC–equity linkage attenuates almost everywhere.** The fully controlled BTC slope $\hat{\beta}_i$ declines in the UK, Euro Area, Japan, China, Brazil, India, and South Africa; the US is essentially flat/slightly higher. Sign stays *positive* throughout.
- **Macro channels sharpen.** The Dollar, rate levels, and curve/slopes absorb more variation post-2019; VIX remains strongly negative; commodity effects become more systematic.

- **Model fit rises post-2019.** Adjusted R^2 increases in most markets (UK is the main exception), consistent with macro controls explaining more of daily equity moves while BTC’s incremental role is smaller.

Key Cross-Period Comparisons

Tables 1–2 condense the *final-column* estimates from each period’s regression panels (daily data). “Pre” and “Post” denote the pre-2019 and post-2019 samples in the provided PDF tables. Coefficients are read as basis points of equity return per 1% BTC move (i.e., $\beta \times 100$ bps).

Table 1: Fully Controlled BTC Coefficient by Country: Pre vs. Post 2019

	United States	United Kingdom	Euro Area	Japan	China	Brazil	India	South Africa
$\hat{\beta}$ (Pre-2019)	0.049	0.039	0.003	0.003	0.011	0.206	0.002	0.045
$\hat{\beta}$ (Post-2019)	0.055	0.016	0.001	0.002	0.001	0.064	0.001	0.013
Sign change?	+→+	+→+	+→+	+→+	+→+	+→+	+→+	+→+
Direction of shift	<i>flat/slightly up</i>	<i>down</i>	<i>down</i>	<i>down</i>	<i>down</i>	<i>down</i>	<i>down</i>	<i>down</i>

Notes: Entries are the BTC slope $\hat{\beta}$ from each period’s most saturated model (BTC + Commodity + Dollar + VIX + short/long + spreads where applicable). Values are expressed as equity bps per 1% BTC move.

Table 2: Adjusted R^2 of Fully Controlled Model: Pre vs. Post 2019

	United States	United Kingdom	Euro Area	Japan	China	Brazil	India	South Africa
Adj. R^2 (Pre-2019)	0.941	0.906	0.915	0.936	0.534	0.815	0.900	0.872
Adj. R^2 (Post-2019)	0.953	0.881	0.937	0.941	0.623	0.821	0.949	0.927

Notes: Values correspond to the final-column specification in each period’s country table.

Country-by-Country Commentary (Fully Controlled Models)

United States. BTC’s partial effect is **essentially unchanged to slightly higher** (0.049→0.055). The Dollar loads more strongly positive post-2019; VIX stays sharply negative. Long rates turn more negative once the front end enters, while short rates are positive—consistent with duration/risk-premium re-pricing dominating at the long end. Fit rises (0.941→0.953). Takeaway: conditional BTC comovement persists, but macro channels carry more of the action.

United Kingdom. BTC falls from 0.039 to 0.016. Commodity remains supportive; Dollar and rate/curve effects strengthen (notably large negatives on the short end and term structure in saturated columns). Adjusted R^2 is slightly lower at the top spec (0.906→0.881), suggesting more idiosyncratic variance in the post period. **Net: conditional BTC linkage weakens materially.**

Euro Area. BTC declines 0.003→0.001. Dollar turns more negative post-2019; 10Y yields are negative when the short end is included, while the front end tends to be positive. Fit improves

(0.915→0.937). Equity sensitivity is increasingly explained by USD and rates; **BTC remains a small but positive partial driver.**

Japan. BTC dips (0.003→0.002). Post-2019 rate/curve terms become very large in absolute value, with long-end negatives and front-end positives once the curve is saturated, plus strongly negative VIX. Fit edges up (0.936→0.941). Interpretation: macro integration tightens; **BTC’s incremental same-day footprint shrinks but stays positive.**

China. BTC drops by an order of magnitude (about 0.011→0.001). Dollar turns more negative; long rates become more negative; slope terms gain importance. Fit increases substantially (0.534→0.623). **The conditional BTC signal is now very small, consistent with stronger macro filtering and local market frictions.**

Brazil. BTC shrinks from a very large effect to a still-meaningful one (0.206→0.064). Dollar becomes more negative; front-end rates turn strongly positive while long-end terms go negative in saturated models. Fit improves (0.815→0.821). **BTC remains a positive factor but no longer dominates once macro channels are controlled.**

India. BTC roughly halves (0.002→0.001). Dollar weakens in the richest models; VIX becomes less negative; 10Y tends to flip negative with the curve in, while 1M stays strongly positive; spreads carry significant explanatory power. Fit rises sharply (0.900→0.949). **Result: conditional BTC elasticity is small but robust.**

South Africa. BTC falls by about two thirds (0.045→0.013). Dollar turns strongly negative post-2019; 10Y is negative and 1Y spread strikingly positive in the saturated model; commodity and front-end rates strengthen. Fit increases (0.872→0.927). **BTC’s positive partial effect persists but is much smaller.**

What the Cross-Period Shift Tells Us

Two robust features emerge. First, the *sign* of BTC’s partial effect on equities remains **positive** across markets *in both periods*, aligning with the model’s parity intuition once BTC is translated into LCU. Second, *magnitudes* generally **decline post-2019**, while the macro block—Dollar, VIX, rates, and term structure—absorbs more variation (higher Adj. R^2). This is consistent with a regime in which global macro forces dominate day-to-day equity moves and the incremental same-day BTC signal is correspondingly smaller, even though it remains directionally aligned with equities.

Partial-Correlation Lens (Formal Note)

Let $\tilde{r}_{i,t}^s$ and $\tilde{r}_{i,t}^k$ denote standardized versions of equity and BTC returns after projecting out X_t . In the fully controlled model,

$$\hat{\beta}_i = \frac{\text{Cov}(r_{i,t}^s, r_{i,t}^k | X_t)}{\text{Var}(r_{i,t}^k | X_t)} \implies \text{pcorr}(r_{i,t}^s, r_{i,t}^k | X_t) = \hat{\beta}_i \cdot \frac{\sigma(r_{i,t}^k | X_t)}{\sigma(r_{i,t}^s | X_t)}.$$

I cannot compute partial correlations **without conditional standard deviations**, but comparing $\hat{\beta}_i$ across periods provides the correct *directional* insight: post-2019 partial comovement is generally **weaker**, not stronger, once macro shocks are held fixed.

Bottom Line

Across eight markets, post-2019 daily equity returns display *less* incremental sensitivity to BTC in LCU, while macro factors explain *more*. The equity–BTC link remains **uniformly positive** (sign prediction of the model), but its economic bite is smaller in the later regime.

Table 3: BTC–Equity Partial Correlations (controlling for macro factors): Pre vs. Post 2019

Country	Pre-2019 r	Post-2019 r	Note
United States	0.404	0.912 ^{LB}	Post uses SE rounded to 0.000; lower bound shown
United Kingdom	0.228	0.543 ^{LB}	Post uses SE rounded to 0.000; lower bound shown
Euro Area	0.108 ^{LB}	0.040 ^{LB}	Both SEs rounded; lower bounds
Japan	0.108 ^{LB}	0.081 ^{LB}	Both SEs rounded; lower bounds
China	0.194	0.040 ^{LB}	Post SE rounded; lower bound
Brazil	0.256	0.791	Exact (SE available in both periods)
India	0.072 ^{LB}	0.040 ^{LB}	Both SEs rounded; lower bounds
South Africa	0.261	0.465 ^{LB}	Post SE rounded; lower bound

Method. For each country and period, I take the BTC coefficient and standard error from the *final* regression column in the PDF tables and compute the partial correlation $r = \sqrt{t^2/(t^2 + \text{df})}$ with $t = \hat{\beta}/\text{SE}$ and $\text{df} = n - p - 1$ (where p counts BTC + Commodity + Dollar + VIX + yield and spread controls used in that column).

^{LB} **Lower bound:** The BTC SE was printed as 0.000 (rounded). I conservatively set $\text{SE}_{\max} = 0.0005$, so $t > \hat{\beta}/0.0005$, yielding a minimum (lower-bound) r . True values are *at least* these.

Data. All inputs come directly from the pre- and post-2019 BTC regression PDFs you provided (daily frequency).

Synthesis. Conditioning on macro factors, BTC’s partial link to equity returns is generally *stronger* after 2019, especially in the U.S. and Brazil; the U.K. and South Africa also rise. Continental Europe and Japan remain low, while China and India are small and roughly flat-to-lower. Because several post-2019 BTC SEs are rounded to 0.000, the reported post-period values in those markets are conservative *lower bounds*; the true partial correlations are at least as large.